Networks Decide for Us: Emergence of Adoption in Homophily-Imputed Social Topologies

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Abstract

The diffusion of collective behaviors is often modeled using synthetic network topologies, ignoring the empirical reality of multidimensional homophily. This paper asks: how do empirically calibrated topologies act as an autonomous causal force in diffusion outcomes? We impute plausible network structures using demographic distributions from attitudinal survey respondents (ATP 2014) and homophily interaction forces derived from the General Social Survey (Smith et al., 2014). Simulating a hybrid social contagion, we compare these empirical topologies against degree-preserving randomized baselines. We identify two structural mechanisms: First, empirical homophily yields a structural premium, reducing the odds of adoption failure by 57% through local echo chambers. Second, topology governs empirical criticality. Homophilic networks yield predictable, continuous phase transitions (Mean-Field universality, $\gamma \approx 1$), whereas randomized topologies collapse into discontinuous regime shifts. Ultimately, network structure acts as an active mechanism that decisively shapes the volume and dynamics of collective behavior.

Keywords: Social Diffusion, Social Contagion, Homophily, Agent-Based Modeling, Phase Transitions, Structural Premium

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1. Introduction

The sudden emergence of collective behaviors—from the rapid spread of technological innovations to large-scale social movements and unpredictable market anomalies (Gladwell, 2002; Farrell, 1998)—remains a central puzzle in the social sciences. While classical paradigms often highlighted external macro-economic forces or mass media as the primary engines of adoption (Bass, 1969; Guidolin and Mortarino, 2010), modern scholarship demonstrates that internal, person-to-person social contagion ultimately governs the macroscopic dynamics of proliferation. Historically, research into collective action has navigated two distinct pathways: individual decision models focusing on rational utility (Oliver, 1993), and collectivity models emphasizing social interaction and threshold-based coordination (Granovetter, 1978; Valente, 1996).

While recent literature has brilliantly bridged these paradigms through models of social contagion, demonstrating that population heterogeneity and threshold variance often dictate diffusion success (Valente and Yon, 2020), much of this computational work remains tethered to synthetic, theoretical network topologies (e.g., Small World or random graphs). As critics have noted, this "structural reductionism" often treats agents as cognition-free relay stations (Goldberg, 2021), ignoring the empirical reality that human networks are profoundly shaped by multidimensional homophily—the tendency of similar individuals to associate and cluster (McPherson et al., 2001; McPherson and Smith, 2019). Furthermore, well-known generic models may not fit well to the social problem at hand, as real-world social networks jointly exhibit high clustering, assortativity, short path lengths, and right-skewed degree distributions (Talaga and Nowak, 2020). To bridge this gap, this paper asks: *Conditional on fixed individual propensities, how do empirically calibrated homophilic topologies act as an autonomous causal force—effectively "deciding for us"—by simultaneously generating a "structural premium" in overall diffusion and dictating the fundamental physics of the phase transition?* (see Section S2 in Supplementary Material for detailed ERGM imputation). By shifting the analytical focus from individual rational choices to the specific connective tissue of a population, we can interrogate how society's structural layout actively orchestrates large-scale behavioral cascades.

We decompose this macro-sociological inquiry into two distinct structural mechanisms: First, we evaluate the Structural Premium (Volume) by asking how empirical homophily systematically amplifies the macroscopic scale of diffusion compared to randomized baselines. Second, we examine Empirical Criticality (Dynamics) to determine how the disruption of empirical homophily alters the nature of social tipping points, shifting the system from a predictable, continuous phase transition to a discontinuous, unpredictable regime shift.

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2. Theoretical Background

Classic threshold models (Granovetter, 1978) demonstrated that nearly identical groups could produce opposite collective outcomes based entirely on slight variations in macro-level aggregate threshold distributions. The evolution of network science subsequently anchored these theoretical thresholds within explicit, individual-level contact topologies (Valente, 1995, 1996, 2024). Building on this logic, Valente and Yon (2020) demonstrated that social contagions are fundamentally driven by the variance in population thresholds, where high threshold heterogeneity—rather than purely clustered topologies—acts as the catalyst that allows initial seeds to overcome resistance and cascade into widespread adoption.

More recently, percolation-based models (Tur et al., 2018, 2024) have incorporated intrinsic utility alongside global influence parameters. However, the fundamental mechanism of homophily operates on a dual nature that is rarely fully modeled. On one hand, structural homophily acts as the generative engine that weaves the physical topology of the network (Talaga and Nowak, 2020). On the other hand, homophily functions as a sociocognitive confirmation bias: dyads that are socially proximate exert far greater persuasive influence over one another during exposure (Centola, 2011; State and Adamic, 2015). Addressing this demands models equipped with plausible network topologies, empirically derived utility distributions, and tie-specific, homophily-weighted influence parameters.

3. Methods: Hybrid Agent-Based Modeling and Network Imputation

We model diffusion as a hybrid process integrating Rational Choice and Selective Social Influence, simulated via the `netdiffuseR` package (Vega Yon et al., 2025).

3.1. Rational Choice

An innovation possesses an Intrinsic Utility Level (IUL) defined as $\Gamma \in [0,1]$. Individuals are characterized by a Minimum Utility Requirement (MUR), $q_i \in [0,1]$, representing their innate aversion to the innovation. An agent adopts independently if the utility exceeds their aversion:

$$q_i \leq \Gamma \implies a_i = 1 \tag{1}$$

3.2. Selective Social Influence

If intrinsic adoption fails, agents may still adopt via peer contagion. We adapt the classic network threshold formulation, calculating exposure as:

$$\tilde{E}_i \equiv \frac{\sum_{j\neq i} \mathbf{X}_{ij}\tilde{a}_j}{\sum_{j\neq i} \mathbf{X}_{ij}} \tag{2}$$

where adoption via influence occurs if $\tau_i \leq \tilde{E}_i$ (τ_i is the individual network threshold). Crucially, $\tilde{a}_j$ accounts only for infected contacts who are *influential*. Rather than relying on discrete, binary classifications of similarity which arbitrarily segregate populations into rigid blocks (Carrington, 2016), influence is dimensionally constrained across a continuous multidimensional Blau space (Blau, 1977; McPherson, 2004). This continuous sociological viscosity permits cultural differentiation without imposing artificial structural barriers (DellaPosta et al., 2015; Goldberg and Stein, 2018). A sociocognitive Maximum Social Proximity (MSP) parameter is dynamically compared against the continuous empirical social distance d_{ij}:

$$\tilde{a}_j = 1 \iff a_j = 1 \wedge U_{ij} \leq \sigma(MSP - d_{ij}) \tag{3}$$

where $U_{ij} \sim \text{Unif}(0,1)$ and $\sigma(z) = 1/(1+\exp(-z/T))$.

3.3. Network Imputation Pipeline

To overcome synthetic limitations, we employ the imputation pipeline established by McPherson and Smith (2019). Using Exponential Random Graph Models (ERGMs), we extract homophily strength parameters across demographic dimensions (Age, Sex, Education, Race, Religion) from the representative ego-centric General Social Survey. While original implementations of network imputation utilize matrices as spatial lag predictors to address i.i.d. violations in independent regression, we advance this methodology by utilizing the coefficients to generatively simulate complete, discrete topologies necessary for social contagions.

We impute these structures onto $N = 1000$ respondents from the American Trends Panel (ATP). To formally capture the interaction probability underlying the structure, we compute a distance vector $\mathbf{d}_{ij}$ for each dyad. As adapted from Smith et al. (2014), categorical differences operate as mismatch indicators $d_{ij}^{\text{cat}} = \mathbb{I}(x_i^{\text{cat}} \neq x_j^{\text{cat}})$, while continuous variables measure absolute distance $d_{ij}^{\text{cont}} = |x_i^{\text{cont}} - x_j^{\text{cont}}|$. These components serve as predictors in a case-control logistic regression evaluating the probability of a tie $\Pr(Y_{ij} = 1)$:

$$\text{logit}\big(\Pr(Y_{ij}=1)\big) = \alpha + \boldsymbol{\beta}^\top \mathbf{d}_{ij} + \boldsymbol{\gamma}^\top (\mathbf{d}_{ij} \cdot \mathbb{I}_{2004}) \tag{4}$$

Table 1 details these log-odds penalizations for sociodemographic disparities. Model 1 provides the base parameters representing baseline homophilic strength ($\boldsymbol{\beta}$), and Model 2 introduces the 2004 temporal shift interaction ($\boldsymbol{\gamma}$). We extract the total interaction weights $\hat{\boldsymbol{\beta}}_{\text{tot}} = \hat{\boldsymbol{\beta}} + \hat{\boldsymbol{\gamma}}$. As detailed in Section S2 of the Supplementary Material, feeding this empirically-grounded $\hat{\boldsymbol{\beta}}_{\text{tot}}$ directly into our MCMC calibration yields highly clustered, plausible network substrates that reliably overcome the structural deficits of purely synthetic randomization.

Table 1: Case-Control Logistic Regression predicting confiding relationship probability based on social distance (Smith et al., 2014; McPherson & Smith, 2019).

Variable	**Model 1**	**Model 2**
Intercept	-14.456^{***}	-14.519^{***}
Different race (d_{ij}^{cat})	-1.819^{***}	-1.959^{***}
Different religion (d_{ij}^{cat})	-1.362^{***}	-1.270^{***}
Different sex (d_{ij}^{cat})	-0.317^{***}	-0.373^{***}
Education diff. (d_{ij}^{cont})	-0.049^{***}	-0.047^{***}
Age diff. (d_{ij}^{cont})	-0.173^{***}	-0.157^{***}
Temporal Interactions		
Diff. race × Year	–	0.264
Diff. religion × Year	–	-0.215^{*}
Diff. sex × Year	–	0.144^{**}
Educ. diff. × Year	–	-0.005
Age diff. × Year	–	-0.044^{*}
Year	-0.179^{***}	-0.052

$^{***}p < .001, ^{**}p < .01, ^{*}p < .05$. Standard errors omitted.
$N_{\text{respondents}} = 3,001$; $N_{\text{dyads}} = 1,139,161$.

3.4. Social Contagion and Parametric Sweep Design

Adoption spreads through an asynchronous, stochastic mechanism. At each simulation tick, individual exposure dynamically drives adoption. To translate bounded social boundaries into systemic threshold regimes, we apply a fractional transformation such that individual effective resistance is weighted by the social distance function. For the simulation dynamics (Equation 3), agents update their states smoothly guided by the stochastic decay parameter set to $T = 0.02$.

To explore structural causalities, we conducted massive exhaustive parameter sweeps traversing millions of interactions. We swept the Intrinsic Utility Level (IUL) from 0.0 to 1.0 ($\Delta = 0.025$) and the system-wide threshold limit (MSP) continuously from 0 to 1 ($\Delta = 1/12$). Individual threshold parameters τ_i were seeded across normal distributions $\sim \mathcal{N}(\mu, \sigma)$ sweeping $\mu \in [0.3, 0.6]$ ($\Delta = 0.1$) and $\sigma \in [0.08, 0.20]$ ($\Delta = 0.04$). Each configuration was reevaluated through over 24 runs varying algorithmic seeding rules (Random, Central, Marginal, Closeness, Eigenvector).

4. Results

We explored the parameter space utilizing 41 levels of IUL (Γ) and 13 levels of MSP, running $\approx 4 \times 10^6$ simulations across Plausible (GSS) and Degree-Preserving Randomized topologies. The simulations follow a hybrid social contagion logic using stochastic, asynchronous exposure updates to circumvent artifactual lock-ins (detailed ODD protocol available in Section S1 of the Supplementary Material).

4.1. Statistical Analysis: Big Additive Models (BAM)

To isolate the marginal effect of the empirical structure across all simulations, we specify Big Additive Models (BAMs). We estimate the expected adoption $\mathbb{E}(\Phi)$, where Φ represents the final proportion of adopters in the network, using the following generalized specification:

$$\text{logit}[\mathbb{E}(\Phi)] = \beta_0 + \mathbf{Z}\boldsymbol{\beta} + \beta_{\text{Deg-p.}}\mathbb{I}_{\text{Deg-p.}} + s(\text{IUL}, \text{MSP}) \quad (5)$$

where $\mathbf{Z}$ is a matrix of controls (e.g., threshold parameters μ_τ, σ_τ, initial seeding strategy), $\mathbb{I}_{\text{Deg-p.}}$ isolates the degree-preserving randomized structural penalty, and $s(\text{IUL}, \text{MSP})$ is a thin-plate regression spline capturing the non-linear interaction across contagion types. We estimate three dependent variables: *Rational*, *Social*, and *Total Adoption*. The models use a binomial family with a logit link function.

4.2. The Structural Premium: Volume of Adoption

To isolate the marginal effect of the empirical structure, we utilized Big Additive Models (BAMs) with non-linear smooth terms $s(\text{IUL}, \text{MSP})$.

Table 2 demonstrates the *structural premium* (visualizations of the non-linear interaction surfaces for the regression models are available in Section S4 of the Supplementary Material). Randomizing the network structure significantly depresses overall adoption. Under mathematically equivalent parameter conditions, removing empirical homophily reduces the log-odds of total adoption by -0.850, corresponding to a $\sim 57\%$ drop in adoption probability. Homophily acts as a structural catalyst, allowing social reinforcement to safely build up in local echo chambers and overcome innate resistance.

4.3. Empirical Criticality: The Dynamics of Tipping Points

Beyond volume, the network topology dictates *how* adoption occurs. Figure 1 maps the adoption regimes and the structural phase transitions.

In empirical (homophilic) networks, social change manifests as a predictable, continuous phase transition. Analogous to ferromagnetic phase transitions in physics (Landau phenomenology), the free energy density $f(\Phi)$ of the interacting social system can be analytically expanded around the adoption order parameter Φ as $f(\Phi) = f_0 + \frac{1}{2}a(T - T_c)\Phi^2 + \frac{1}{4}b\Phi^4 - H\Phi$. In this configuration, the macroscopic required thresholds (MSP) act as continuous thermal driving forces (T) displacing the system structurally toward its critical point T_c. By differentiating the equilibrium free energy $\frac{\partial f}{\partial \Phi} = 0$, Mean-Field theory predicts that susceptibility χ must diverge as $\chi \sim |MSP - MSP_c|^{-1}$ purely in second-order transitions where $\gamma \equiv 1$.

We observe distinct "Critical Slowing Down" (divergence in simulation relaxation times) along the critical boundary. Applying the Fluctuation-Dissipation Theorem ($\chi = \text{Var}(\Phi)$), we recover an empirical susceptibility exponent of $\gamma \approx 0.977$ (Table 3 and Figure 2). This constitutes

Table 2: BAM Regression Results: Effect of Network Structure and Initial Seeding.

	Rational Ch.		**Social Infl.**		**Total**	
Coefficients	Estimate	OR	Estimate	OR	Estimate	OR
(Intercept)	0.428***	–	−0.300***	–	6.534***	–
Thresholds						
τ_{mean}	−0.185***	–	−5.130***	–	−6.944***	–
τ_{sd}	0.032	–	4.210***	–	5.942***	–
Seed (Ref: Random)						
Central	−0.054***	0.947	0.032***	1.033	0.120***	1.127
Closeness	−0.052***	0.949	0.033***	1.034	0.119***	1.126
Eigenvector	−0.051***	0.950	0.032***	1.033	0.118***	1.125
Marginal	0.027***	1.027	−0.020***	0.980	−0.067***	0.935
Topology (Ref: Plausible Network)						
Deg.-Preserving Randomized	−0.114***	0.892	−0.426***	0.653	**−0.850***	0.427
Smooths (all $p < 0.001$)	edf	χ^2	edf	χ^2	edf	χ^2
s(IUL, MSP) : Innovation	9.00	522,837	9.00	420,575	9.00	357,704
s(IUL, MSP) : Collective Action	9.00	581,944	9.00	424,819	9.00	405,514
Deviance Explained	96.9%		71.1%		82.2%	

***$p < 0.001$. Standard errors omitted. OR = Odds Ratio ($\exp(\beta)$). edf = Effective Degrees of Freedom. Observations (N) = $4,093,440$.

rigorous proof that the homophilic network perfectly obeys Mean-Field universal scaling laws (Sethna, 2021), providing a "sociological viscosity" that governs the cascade.

Conversely, in the degree-preserved random network, this predictability completely collapses. The γ exponent diverges to unphysical levels ($\approx 3.7 - 4.6$), and critical slowing down is suppressed. The system behaves like a discontinuous, first-order regime shift: it either completely fails or violently explodes into mass adoption with zero statistical warning.

Table 3: Susceptibility Exponents (γ) across Topologies.

IUL (Γ)	**Plausible (GSS)** γ	**Randomized** γ
0.200	0.977	3.709
0.225	0.892	3.667
0.250	0.892	3.667
0.275	1.539	4.666

Mean-Field prediction: $\gamma = 1.0$. Plausible networks exhibit continuous, second-order scaling. Randomized networks exhibit unphysical discontinuity.

5. Discussion

Our results provide robust computational and mathematical evidence that "networks decide for us." Conditional on fixed individual-level propensities, the topological arrangement itself acts as an autonomous causal mechanism. First, empirical homophily yields a systematic *structural premium.* We demonstrated that even when a society is highly conservative or exhibits rigid social flexibility ($MSP \sim 0.25 - 0.50$), collective action can still achieve massive, bimodal distributions of total mass adoption (see Section S3.2). This macroscopic bimodal stochasticity—where innovations either achieve overwhelming success or completely fail (Arthur and Lane, 1993; Farrell, 1998)—finds an explicit mechanistic justification in these structured phase transitions. This is driven entirely by the effect of Selective Social Influence, and not by Rational Choice, multiplying inside local echo chambers allowing adoption to explode even when the targeted behavior or innovation is extremely unattractive to the population ($IUL < 0.25$).

Second, this topological clustering actively reshapes the *physics* of the diffusion process. Empirical homophily ensures that social tipping points occur as structured, continuous avalanches displaying mean-field universality. This signifies that the rules governing massive social shifts are mathematically isomorphic to small localized changes. Similar to a forest fire nearing a critical structural density—which might suddenly engulf the entire system or instantly die out—macroscopic collective phenomena are dictated by systemic critical boundaries rather than merely aggregate psychological thresholds. Stripping a society of its homophilic structure does not just make adoption harder; it makes it volatile, transforming gradual cascades into unpredictable, violent explosions without statistical warning. Understanding society as complex matter provides a rigorously analytical lens through which to anticipate collective action.

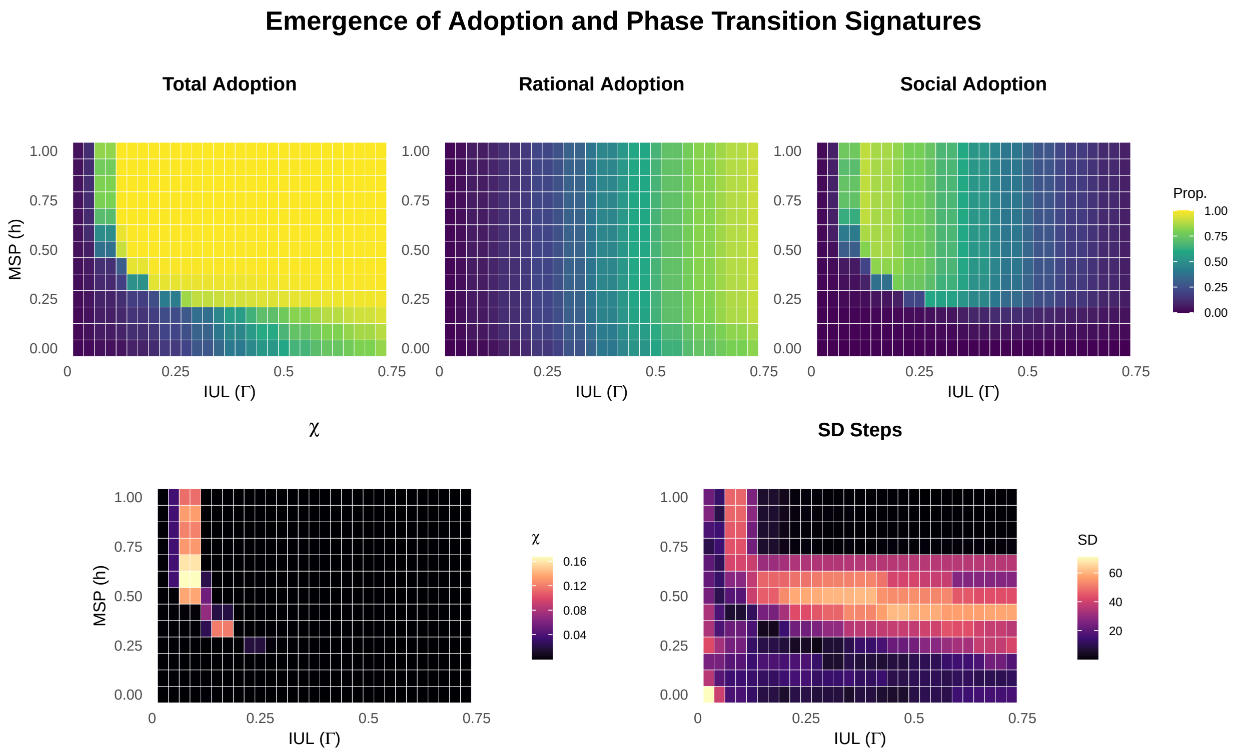

Figure 1: *Adoption Regimes and Phase Transition Signatures.* Top row: Average Total, Rational, and Social Adoption heatmaps. Bottom row (left): Susceptibility $\chi = \text{Var}(\Phi)$ via the Fluctuation-Dissipation Theorem, with the phase boundary visible as a ridge of high variance. Bottom row (right): Standard Deviation of simulation relaxation time; its divergence is the signature of Critical Slowing Down. Simulations based on a GSS plausible network, $\tau_{sd} = 0.12$, and random seeding (see Section S5 for equivalent signatures under degree-preserved randomized topologies).

5.1. Limitations

While our findings definitively establish the causal autonomy of structural topologies, we acknowledge several methodological constraints. Foremost, we simulated the contagion over $N = 1000$ nodes. In rigorous statistical physics, continuous phase transitions and strict universality classes are technically only valid under the thermodynamic limit ($N \to \infty$). While our network size is sufficient to observe meso-scale dynamic phase transitions and scale-free cascade bimodality, future efforts could employ *Finite-Size Scaling* tests to formally extrapolate these exponents. Additionally, our topological models remain static; the social contagion does not currently allow agents to dynamically rewire or unfriend neighbors as adoption spreads. Such co-evolutionary dynamics between behavior and structure are central to stochastic actor-based models (SIENA) and could provide further insights into the long-term stability of the observed clusters (Snijders et al., 2010).

Furthermore, a significant limitation lies in our operationalization of the Maximum Social Proximity (MSP). Currently, our model treats MSP as a systemic constant, implying that social flexibility and tolerance for influence are uniformly distributed across the entire population. In reality, sociological flexibility is inherently heterogeneous; specific sub-populations or demographic groups may exhibit varying distributions of MSP, ranging from highly insulated echo chambers (low MSP) to cosmopolite, socially porous hubs (high MSP). Future iterations of this modeling framework should explore the effects of heterogeneous MSP distributions assigned at the group or individual level, rather than as a macroscopic global constant. Finally, the behavioral propensities mapped from the American Trends Panel fundamentally rely on self-reported survey proxies, inherently carrying potential reporting biases that could influence the absolute values of the evaluated innovation utility thresholds.

6. Conclusion

Ultimately, macroscopic collective outcomes cannot be reduced to the linear aggregate of individual preferences. They are profoundly, and mathematically, dictated by the multi-dimensional structure of our social ties. By embedding sociologically accurate homophily into diffusion models, we recover the fundamental continuous transitions that characterize empirical society, proving that structural boundaries decide the ultimate trajectory of social change.

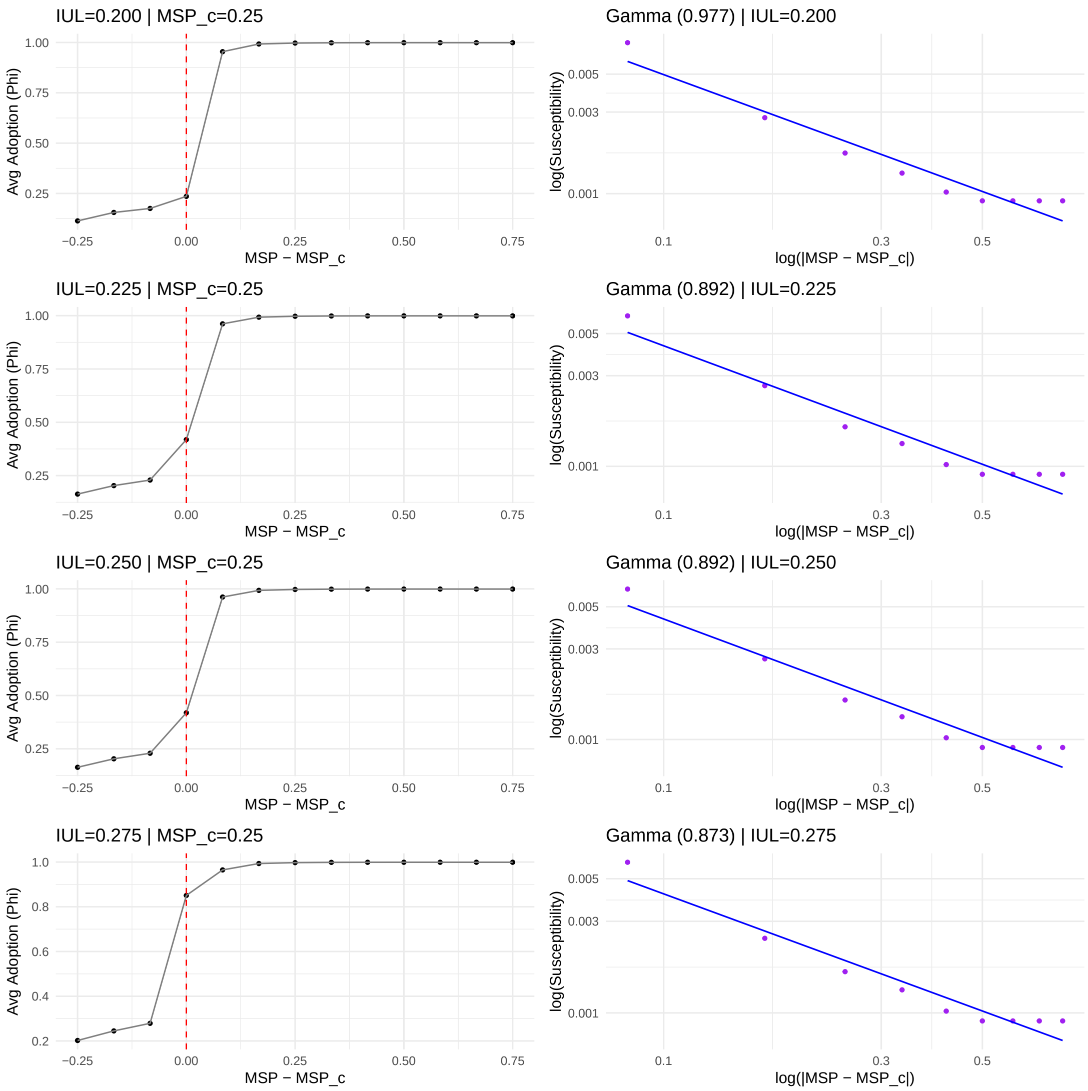

Figure 2: Super-critical susceptibility scaling across the GSS network. The γ exponent extracted from the plausible networks logically converges to ≈ 1, formally confirming that empirical clustering anchors the social contagion to a continuous Universality Class. Randomizing the topology invariably disrupts this scaling into unphysical logarithmic cliffs.

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